LesionNet: Lesion-Gated Classification for Background-Robust, Faithfully Explainable and Severity-Aware Plant Disease Diagnosis

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***Abstract*—Convolutional neural networks (CNNs) reach near-perfect accuracy on the PlantVillage benchmark, yet it is unclear whether they recognise the disease or the photographic setting in which each class was captured. We first show, on a held-out split of 3,109 images from 15 tomato, potato and pepper classes, that four fine-tuned CNNs place only 7–12% of their Grad-CAM++ evidence on diseased tissue and 25–49% on the background, and that the most accurate of them (ResNet50, 99.8%) falls to 45.1% when the same leaves are placed on a plain grey background. We then propose *LesionNet*, a segment-then-classify network in which a segmentation head labels every pixel as background, healthy leaf or lesion, and the classifier receives only features pooled over the predicted lesion and leaf pixels plus the predicted lesion area (*lesion-gated pooling*). The gate is detached from the classification loss, so the lesion map is not a post-hoc explanation but the actual input to the decision, and it doubles as a severity estimate. LesionNet is trained without any manual annotation, from colour-based pseudo-masks and background-swap augmentation. With an EfficientNetB0 backbone it matches the baselines on clean images (99.5%) while keeping 98.7% accuracy on grey backgrounds, against 67.5% for an identically trained plain EfficientNetB0. Its evidence lands on a lesion in 97% of images (CNNs: 49–64%), and its predicted diseased area correlates with the measured area at Spearman $\rho = 0.90$. A cross-dataset test on 1,012 PlantDoc field photographs shows that all models, LesionNet included, drop to 19.6–23.6%, indicating that background robustness is necessary but not sufficient for field deployment. LesionNet is integrated into AGRI-X, an open-source decision-support application for farmers.**

***Index Terms*—plant disease detection, shortcut learning, explainable AI, interpretable-by-design, weak supervision, lesion segmentation, severity estimation, PlantVillage, PlantDoc**

I. Introduction

Crop diseases cause large yield losses, and in many regions a farmer's first point of diagnosis is a mobile phone photograph. Since the release of PlantVillage [1], transfer-learned CNNs have repeatedly reported 97–99% accuracy on leaf images [2]–[4]. These figures have not translated into equally reliable tools in the field [5], [6]: PlantVillage images were captured under controlled conditions in which the background, lighting and capture session differ between classes, and a model can exploit these cues [7]. This is an instance of *shortcut learning* [8], which background-sensitivity tests have exposed in general object recognition as well [9].

Two practical requirements follow. First, a diagnostic model should be *right for the right reasons* [10]: its prediction should depend on the diseased tissue and not on the table the leaf was photographed on. Second, the explanation shown to the user should be *faithful*. Post-hoc saliency maps such as Grad-CAM [11] and Grad-CAM++ [12] approximate where a network looked, can be insensitive to the model they explain [13], and are coarse (7×7 for the backbones studied here). For high-stakes decisions an interpretable-by-design model is preferable to an explained black box [14]. A third, agronomic requirement is *severity*: the treatment decision depends on how much of the leaf is affected, which a class label does not provide [15].

We address all three with a single architectural constraint: *the classifier may only see what the network has outlined as leaf or lesion.* Our contributions are:

1) **Diagnostic study.** On a held-out test split we quantify how four standard CNNs (MobileNetV2, ResNet50, EfficientNetB0, DenseNet121) distribute their evidence and how they behave under controlled background interventions (grey, swapped, leaf-removed). The most accurate model is the least robust. We also document a Grad-CAM implementation pitfall in Keras 3 that silently corrupts explanations of nested models.
2) **LesionNet.** A segment-then-classify network with *lesion-gated pooling*: class features are pooled only over predicted lesion and leaf pixels, with a stop-gradient gate so that the classification loss cannot move the gate towards convenient background regions. The lesion map is, by construction, the evidence the decision is computed from.
3) **Annotation-free training.** LesionNet is trained from colour-based pseudo-masks with healthy-class negative supervision and reliability weighting, plus background-swap augmentation. No pixel is labelled by hand.
4) **Robustness, faithfulness and severity.** LesionNet matches the best CNN on clean images, reduces errors on grey backgrounds by $25\times$ relative to an identically trained plain backbone, and yields a lesion-area severity estimate with $\rho = 0.90$, where Grad-CAM areas show no correlation ($|\rho| \leq 0.13$).
5) **Honest cross-dataset evaluation and deployment.** On PlantDoc field photographs [16] every PlantVillage-

trained model, LesionNet included, collapses to 19.6–23.6% accuracy, a negative result we report in full. Code, results and the AGRI-X application are released openly.

II. Related Work

Plant disease classification. Mohanty *et al.* [2] reported 99.35% accuracy on PlantVillage with transfer learning; Ferentinos [3] and Too *et al.* [4] compared deeper architectures with similar results. Barbedo [5], [6] identified limited dataset variety and capture conditions as the main obstacles to field use. PlantDoc [16] provided internet-sourced field images and showed large accuracy drops for models trained on lab images.

Dataset bias and shortcut learning. Noyan [7] showed that PlantVillage classes can be predicted far above chance from background pixels alone. Geirhos *et al.* [8] survey shortcut learning, and Xiao *et al.* [9] introduced background-swap tests for ImageNet models. We adapt such interventions to leaf images using automatic leaf masks and add a "background only" variant without the leaf silhouette.

Explanations and interpretable-by-design models. Grad-CAM [11] and Grad-CAM++ [12] are the standard explanations in plant pathology papers, but saliency maps can fail basic sanity checks [13]. The pointing game [17] is a common localisation metric. Interpretable-by-design alternatives include prototype networks [18] and attention branches [19]; in the latter, attention re-weights features but background features still reach the classifier. Ross *et al.* [10] penalise input gradients outside annotated regions, which requires masks at training time. LesionNet instead removes background features from the classifier input by construction and obtains its masks automatically.

Lesion segmentation and severity. Wang *et al.* [15] classified severity stages directly from images. Pixel-level lesion datasets such as PlantSeg [20] are recent and costly to build. Our pseudo-masks follow the pseudo-labelling idea [21], generated by a colour model with GrabCut refinement [22] instead of a teacher network.

III. Method

A. Problem setting

Given an RGB leaf image x, the model predicts one of $K{=}15$ classes (Table I). Each training image has a class label y and an automatically generated label map $m \in \{0,1,2\}^{H\times W}$ (background, healthy leaf, lesion) with a reliability weight $w \in \{0,1\}$. No manual pixel labels are used anywhere.

B. Annotation-free pseudo-masks

Pseudo-masks are produced by a deterministic colour pipeline, developed for the AGRI-X application and frozen before any LesionNet experiment.

1) *Leaf.* A background colour model is estimated from a band along the image border, excluding clearly green pixels. Plant-coloured pixels (green, yellow-green, brown with sufficient saturation) seed the leaf; pale leaves are recovered by an Otsu threshold on greenness ($-a^*$ in CIELAB). The coarse mask is refined with GrabCut [22]. Dark necrotic tissue that touches the leaf and differs in chroma from the background is added back, the largest component is kept and holes are filled.
2) *Healthy reference.* The healthy colour of *this* leaf is the median (V, S, a^*, b^*) of clearly green, well-lit pixels (HSV hue 34–88, $S, V \geq 50$), inside the leaf minus a 2.5% rim.
3) *Lesions.* Pixels are necrotic if they are brown or red-brown and darker than the reference, very dark, or shifted by more than 14 units towards red in a^*. They are chlorotic if they are yellow or measurably yellower than the reference hue. For the two classes with white fungal growth (late blight, leaf mould) a pale class is added.
4) *Clean-up.* Morphological opening and closing remove specks. Components smaller than 0.08% of the leaf are discarded, as are strips along the leaf outline whose colour matches the adjacent surface (cast shadows, background bleed).

For the 3,221 healthy-class images every lesion pixel is relabelled as healthy leaf, which yields clean negative examples. A diseased image receives $w{=}1$ only if the pipeline's own reliability checks pass (the leaf covers 12–97% of the frame and enough healthy tissue remains as reference) and at least 1% of the leaf is lesion. Otherwise $w{=}0$ and the image trains the classifier only, so the network is never told that a sick leaf is healthy. Of 20,638 images, 17,895 (86.7%) receive segmentation supervision. Lesions cover a median of 8.8% of the leaf (mean 12.8%), and the leaf covers on average 42.5% of the image.

C. Architecture

LesionNet (Fig. 1) uses an ImageNet-pretrained [23] backbone, EfficientNetB0 [24] by default or MobileNetV2 [25], and taps features at strides 8, 16 and 32. A light FPN-style decoder [26] (1×1 lateral convolutions to 128 channels, bilinear top-down addition, two 3×3 conv-BN-ReLU blocks and a 2× upsampling) outputs class probabilities $P \in [0,1]^{56\times56\times3}$. The class branch concatenates stride-16 features with upsampled stride-32 features and projects them to $F \in \mathbb{R}^{14\times14\times256}$.

D. Lesion-gated pooling

Let $\tilde{P}$ be P area-averaged to 14×14 (4×4 average pooling) and passed through a stop-gradient $\mathrm{sg}(\cdot)$. With lesion gate $g_\ell = \tilde{P}_{\text{lesion}}$ and leaf gate $g_f = \tilde{P}_{\text{leaf}} + \tilde{P}_{\text{lesion}}$,

$$\bar{f}_\ell = \frac{\sum_u g_\ell(u)\, F(u)}{\sum_u g_\ell(u) + \epsilon}, \qquad \bar{f}_f = \frac{\sum_u g_f(u)\, F(u)}{\sum_u g_f(u) + \epsilon}, \quad (1)$$

$$a = \frac{\sum_u g_\ell(u)}{\sum_u g_f(u) + \epsilon}, \qquad z = [\, \bar{f}_\ell \,\|\, \bar{f}_f \,\|\, a \,] \in \mathbb{R}^{513}, \quad (2)$$

with $\epsilon = 10^{-3}$. The vector z passes through dropout (0.3), a 256-unit ReLU layer and a softmax. Three properties follow.

- *Gated input.* Positions that the segmentation head assigns to background ($g_f(u) \approx 0$) contribute nothing to z, so the decision is computed from leaf and lesion positions only. Features at those positions still have receptive fields

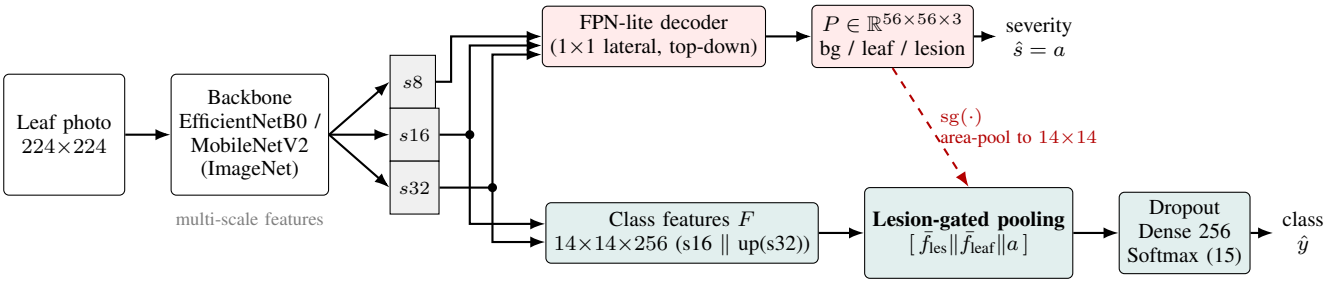


Fig. 1. LesionNet. A segmentation head labels each pixel as background, healthy leaf or lesion. The classifier sees only features averaged over predicted lesion pixels ($\bar{f}_{\text{les}}$) and leaf pixels ($\bar{f}_{\text{leaf}}$), plus the lesion-to-leaf area ratio a. The gate is detached (sg = stop-gradient), so only the pseudo-masks shape it. The same map is shown to the user and gives the severity $\hat{s}$.

that extend past the leaf boundary; we return to this in Sec. VI.

- *Faithful explanation.* The map shown to the user, P_{lesion}, is the weighting used by the classifier, not an approximation computed afterwards [14].
- *Severity for free.* a is the predicted diseased fraction of the leaf. It is an input to the classifier and is reported to the user as severity (computed at 56×56).

The stop-gradient is a deliberate design choice. Without it, the classification loss could enlarge the lesion gate over any region that is discriminative, background included, and the map would drift away from disease tissue.

E. Training objective and augmentation

The loss is

$$\mathcal{L} = \mathrm{CE}_{\kappa}(y, \hat{y}) + \lambda\, w\, \frac{1}{|\Omega|} \sum_{u\in\Omega} \beta_{m(u)}\, \mathrm{CE}\big(m(u), P(u)\big), \quad (3)$$

where CE_{κ} is cross-entropy with class weights $\kappa_c = \mathrm{clip}(N/(KN_c), 0.5, 3)$ to offset class imbalance, $\lambda = 1$, $\beta_{\text{lesion}} = 3$ and $\beta = 1$ otherwise (lesion pixels are rare).

All augmentations are applied jointly to image and mask: random horizontal and vertical flips, 90° rotations, and brightness (±0.12), contrast and saturation ($\times[0.85, 1.15]$) and hue (±0.02) jitter. With probability 0.5 every pixel labelled background is replaced by a synthetic surface (random colour, linear gradient and Gaussian noise). This *background swap* removes the correlation between surface and class during training.

IV. Experimental Setup

A. Data

In-domain. We use the 15 pepper, potato and tomato classes of PlantVillage [1] (Table I), resized to 224×224. A stratified 70/15/15 split with a fixed seed is shared by all LesionNet variants and by the evaluation.

Cross-dataset. PlantDoc [16] contains field and internet photographs. Twelve of its classes correspond to ours (it has no potato-healthy, target-spot or spider-mite class). We downloaded all images of these classes from the public repository (train and test folders). 1,012 were readable, and they are used *only for testing*. Images are resized to 224×224 without cropping. Chance is 8.3% for this 12-way task.

TABLE I
DATASET: THE 15 PLANTVILLAGE CLASSES USED (20,638 IMAGES), SPLIT 70/15/15 PER CLASS (SEED 42) INTO 14,440 / 3,089 / 3,109 IMAGES.

Crop	Class	n	Crop	Class	n
Pepper	Bacterial spot	997	Tomato	Bacterial spot	2127
Pepper	Healthy	1478	Tomato	Early blight	1000
Potato	Early blight	1000	Tomato	Late blight	1909
Potato	Late blight	1000	Tomato	Leaf mould	952
Potato	Healthy	152	Tomato	Septoria spot	1771
Tomato	Healthy	1591	Tomato	Spider mites	1676
			Tomato	Target spot	1404
			Tomato	Yellow leaf curl virus	3208
			Tomato	Mosaic virus	373

B. Models

Baselines. MobileNetV2 [25], ResNet50 [27], EfficientNetB0 [24] and DenseNet121 [28], each ImageNet-pretrained with a GAP–BN–dense head, trained in two phases: 5 epochs with a frozen backbone (lr 10^{-4}), then 15 epochs fine-tuning the last 50 layers (lr 10^{-5}). They were trained earlier with their own 70/15/15 split, so some of our test images may have been in their training data. This can only *inflate* their clean accuracy.

LesionNet variants. Trained identically on our split: 3 epochs with the backbone frozen (Adam [29], lr 10^{-3}), then 15 epochs end-to-end with cosine decay [30] from 10^{-4} to 10^{-6}, batch 32, BatchNorm statistics frozen, mixed precision [31], early stopping (patience 5) and checkpointing on validation accuracy.

- **A3, LesionNet**: EfficientNetB0 backbone with all components.
- **A0, Plain-EffNet**: the same network with global average pooling instead of the gate, λ=0 and no background swap. It is a like-for-like baseline: same backbone, split, schedule and training code.
- **A5, LesionNet-Mobile**: MobileNetV2 backbone. A GPU driver fault stopped its training after epoch 10 of 18; the best checkpoint (validation 96.2%) is used.

Training used one NVIDIA RTX 4050 Laptop GPU (6 GB) under WSL2 with TensorFlow/Keras 3. A3 trained in 76 minutes and A0 in 42 minutes.

TABLE II
ACCURACY (%) ON THE PLANTVILLAGE TEST SPLIT (3,109 IMAGES) UNDER BACKGROUND INTERVENTIONS. BG.-ONLY CHANCE: 6.7%. SHADED: OURS. BEST VALUES IN BOLD.

Model	Clean	F1	Grey	Swapped	Silh.↓	Bg. only↓
MobileNetV2	93.0	.919	76.0	71.6	25.5	19.1
ResNet50	**99.8**	**.998**	45.1	60.0	33.6	37.4
EfficientNetB0	96.4	.963	81.2	62.3	26.2	**17.8**
DenseNet121	96.2	.962	56.4	57.6	30.1	28.9
A0 Plain-EffNet	99.6	.997	67.5	63.7	47.2	32.7
A5 LesionNet-Mobile	96.0	.959	94.2	73.2	26.5	20.3
A3 LesionNet	99.5	.995	**98.7**	**81.8**	27.2	19.5

TABLE III
FAITHFULNESS OF THE EVIDENCE ON 2,222 DISEASED TEST IMAGES, AND SEVERITY. MAP: GRAD-CAM++ FOR CNNS, OWN LESION MAP FOR LESIONNET. A0 IS OMITTED (ITS SEGMENTATION HEAD IS UNTRAINED).

Model	Lesion↑	Bg.↓	Focus↑	Point.↑	IoU↑	Sev. ρ ↑
MobileNetV2	.094	.288	0.96	.492	.109	–
ResNet50	.087	.381	1.06	.636	.117	–
EfficientNetB0	.121	.247	1.22	.630	.133	–
DenseNet121	.071	.487	1.03	.604	.081	–
A5 LesionNet-Mobile	.340	.082	4.12	.949	.410	.784
A3 LesionNet	**.441**	**.067**	**6.06**	**.971**	**.505**	**.900**

C. Background interventions

Using the pseudo leaf mask of each test image we build five variants (Fig. 2):

- *clean*;
- *grey*: background set to 50% grey;
- *swapped*: the leaf pasted onto the full photograph of a test image from a *different* class, which may include part of that image's leaf;
- *silhouette*: leaf set to grey, background kept;
- *background only*: leaf pixels filled with resampled and blurred pixels of the image's own background, so neither leaf nor silhouette remains.

A model that relies on the leaf should keep its accuracy on grey and swapped images and fall to chance (6.7%) on background-only images.

D. Faithfulness and severity metrics

CNNs are explained with Grad-CAM++ for the predicted class at their last convolutional block, upsampled bilinearly to 224×224; LesionNet with P_{lesion}. On the 2,222 diseased test images with a reliable pseudo-mask containing at least 1% lesion, we report:

- the share of map mass on lesion and on background pixels;
- the *focus ratio*, i.e. mean map value on lesions divided by mean value on the whole leaf (1 = no preference);
- the *pointing game* [17]: the map's maximum lies within 15 px of a lesion pixel;
- the IoU between the map thresholded at 0.5 and the lesion mask.

Severity is evaluated with Spearman's ρ and the mean absolute error (MAE) between predicted and pseudo-mask lesion percentage, plus the mean predicted lesion percentage on the 487 healthy test images. Accuracy 95% confidence intervals use the Wilson score [32]; for $n = 3{,}109$ they are at most ±1.8 points.

V. RESULTS

A. Accuracy under background interventions

Table II and Fig. 3 summarise the main result. On clean images every model except MobileNetV2 exceeds 96%, and the ranking by clean accuracy (ResNet50 first) is close to the *reverse* of the ranking by robustness. ResNet50 loses 54.7 points when the background becomes grey, and it identifies the class from background pixels alone in 37.4% of images, 5.6× chance. All four CNNs remain far above chance on background-only images (17.8–37.4%), confirming a substantial background shortcut on this benchmark [7].

LesionNet is as accurate as the best CNN on clean images (99.5% vs. 99.8%, overlapping confidence intervals) and loses only 0.8 points on grey backgrounds (98.7%, CI 98.3–99.1). Its grey-background error is 1.3%, against 18.8% for the most robust CNN (EfficientNetB0). On the harder swapped test, where another leaf can be visible in the pasted background, it reaches 81.8% against 57.6–71.6% for the CNNs.

B. Ablation: where does the robustness come from?

A0 shares LesionNet's backbone, split, schedule and code, and is equally accurate on clean images (99.6%). It nevertheless reaches only 67.5% on grey backgrounds (31.2 points below LesionNet, 25× the error), 63.7% on swapped ones, and 32.7% from the background alone, against LesionNet's 19.5%. Its silhouette accuracy (47.2%) shows that it also exploits leaf shape, which identifies the crop. The robustness is therefore produced by the LesionNet components, not by the backbone or the training recipe. The same holds for the light backbone: LesionNet-Mobile keeps 94.2% on grey backgrounds, versus 76.0% for the CNN MobileNetV2, at similar clean accuracy (96.0% vs. 93.0%).

The remaining ablations (background swap only; segmentation loss with global pooling; gate without stop-gradient) are implemented in the released code but were not run for this paper.

C. Does the evidence lie on the disease?

Table III and Fig. 4 show that the CNN explanations are nearly indifferent to disease. Their focus ratios of 0.96–1.22 mean that a lesion pixel receives about as much attention as a healthy one, and 25–49% of the map mass falls on the background. Their maps' peaks land on a lesion in 49–64% of images. In our earlier 600-image study, a random point on the leaf hit a lesion 53% of the time, so these peaks are barely better than chance. LesionNet places 44% of its evidence on lesions and 6.7% on background, its peak is on a lesion in 97% of images, and its map overlaps the lesion mask at IoU 0.50, against 0.08–0.13 for the CNNs.



Fig. 2. Background interventions on a test image (tomato early blight). (b) Automatic pseudo-mask: green = healthy leaf, red = lesion. (c) Leaf on 50% grey. (d) Leaf pasted onto a photo of another class (pepper, healthy). (e) Leaf removed, silhouette kept. (f) Leaf filled with the image's own background texture.

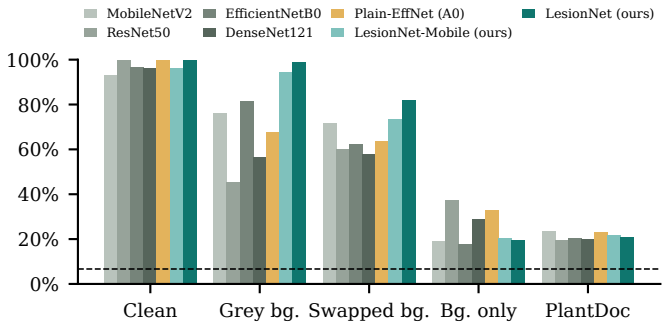


Fig. 3. Accuracy under interventions and on PlantDoc field photos. Dashed line: 15-way chance (6.7%); PlantDoc chance is 8.3%. For "Bg. only", lower is better.

Caveat. These metrics use the same automatic masks that LesionNet learns from, which favours it. The intervention tests of Table II and the cross-dataset test use no masks at test time (only the leaf mask needed to build the variants) and are the unbiased comparisons.

D. Severity estimation

LesionNet's predicted lesion fraction correlates with the measured fraction at $\rho = 0.900$ (MAE 5.7 percentage points). On healthy leaves it predicts 0.55% diseased area on average. LesionNet-Mobile reaches $\rho = 0.784$ (MAE 10.3, 1.3% on healthy leaves). For comparison, in our earlier 600-image study the "affected area" derived from Grad-CAM (share of the leaf with attention above 0.25), as used in many applications, showed no relationship with lesion area ($|\rho| \leq 0.13$ for all four CNNs) and reported 66–100% affected area on perfectly healthy leaves.

E. Cross-dataset generalisation (negative result)

Table IV shows that every model, LesionNet included, generalises poorly to field images: 19.6–23.6% accuracy on 12 classes. The differences between models fall within the confidence interval (about ± 2.6 points). Models identify the crop in 62–71% of images but rarely the disease. Fig. 5 illustrates why. Field images contain several overlapping leaves, soil, hands, specular highlights and blur. LesionNet's segmentation, trained on single leaves on uniform surfaces, marks sunlit soil and out-of-focus leaves as lesion; on PlantDoc it predicts 16.5% lesion on healthy leaves, against 0.55% on PlantVillage. Removing the background shortcut is therefore necessary but not sufficient: the remaining gap is a domain shift in the leaf and lesion appearance itself.

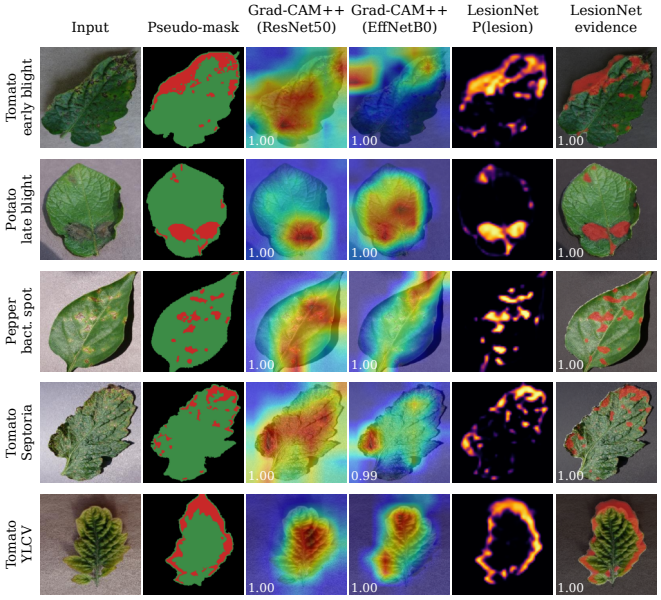


Fig. 4. Test images. Grad-CAM++ maps of the CNNs are coarse and often centred on healthy tissue or background. LesionNet's lesion probability and the evidence its decision is computed from follow individual spots. Numbers: predicted-class confidence (all predictions correct). Last row: the lesion map of yellow leaf curl virus follows the chlorotic margins.

TABLE IV
CROSS-DATASET TEST ON 1,012 PLANTDOC FIELD IMAGES (12 SHARED CLASSES, CHANCE 8.3%). NO MODEL SAW PLANTDOC DURING TRAINING. 95% CI $\approx \pm 2.6$ POINTS.

Model	Accuracy (%)	Macro-F1	Correct crop (%)
MobileNetV2	**23.6**	.189	68.9
ResNet50	19.6	.147	65.2
EfficientNetB0	20.2	.148	68.9
DenseNet121	20.1	.158	67.4
A0 Plain-EffNet	23.1	**.191**	70.4
A5 LesionNet-Mobile	21.6	.154	62.5
A3 LesionNet	20.8	.172	**70.7**

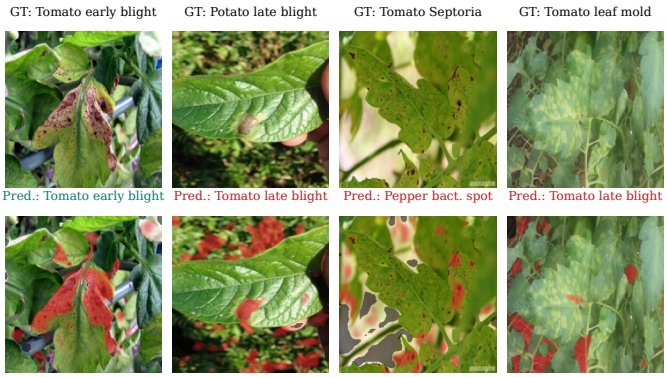


Fig. 5. LesionNet on PlantDoc field images (top: input with ground truth; bottom: lesion evidence and prediction). Single dominant leaves with clear lesions are handled (left); sunlit soil, overlapping leaves and blur are mistaken for lesions (right).

TABLE V
MODEL COST. PARAMETERS; BATCH-1 FLOAT32 LATENCY ON A 2-CORE X86 CPU (MEDIAN OF 20 RUNS, TENSORFLOW GRAPH MODE).

Model	Params (M)	CPU latency (ms)
MobileNetV2	2.63	26.2
ResNet50	24.78	96.8
EfficientNetB0	4.39	50.3
DenseNet121	7.34	75.9
A5 LesionNet-Mobile	3.35	33.1
A3 LesionNet	4.95	50.7

F. Computational cost

The segmentation decoder and gated head add 0.56 M parameters to EfficientNetB0 (+13%) and, in our measurement, no measurable latency (50.7 vs. 50.3 ms, Table V). LesionNet-Mobile runs in 33 ms on a CPU, so on-device use is feasible. LesionNet uses 20% of ResNet50's parameters and half its latency while being far more robust.

G. Implementation pitfall: Grad-CAM on nested Keras 3 models

In Keras 3, applying a backbone's `preprocess_input` to a symbolic tensor creates graph operations, not layers. A common Grad-CAM implementation rebuilds the sub-graph before the backbone from `model.layers` and therefore skips this preprocessing. In our original pipeline the gradient model chose the same class as the actual model in only 22/60 (ResNet50) and 4/60 (DenseNet121) images, and the heatmap correlation fell to 0.15 for DenseNet121. EfficientNetB0 and MobileNetV2 were unaffected because their preprocessing is internal or absent. All results in this paper use a corrected builder that reproduces each model's softmax exactly (maximum difference 0.0). We recommend this check whenever Grad-CAM is used to compare backbones.

VI. DISCUSSION

Why the gate helps. A global-pooling classifier is free to weight any spatial position. When surfaces differ by class, the background is the easiest feature to learn. Lesion-gated pooling removes background positions from the classifier input, and the detached gate prevents the classifier from widening the gate to recover them. Background swap complements this by breaking the surface–class correlation at the segmentation stage.

Why some background signal remains. LesionNet still classifies 19.5% of background-only images correctly, against a chance level of 6.7%. Two mechanisms are plausible. (i) Features at leaf positions have receptive fields that cover neighbouring background, so background context leaks into F near the leaf boundary. (ii) On background-only images the segmentation head may label some background texture as leaf. Shrinking the receptive field near the boundary, or erasing background before the backbone, are natural next steps; both trade context for invariance.

Accuracy is not evidence of understanding. The most accurate model in our study (ResNet50, 99.8%) is the least robust (45.1% on grey backgrounds). Benchmarks on PlantVillage should report intervention accuracies alongside clean accuracy; the protocol is cheap because it needs only automatic leaf masks.

Limitations and threats to validity.

1) The pseudo-masks are colour-based. They under-detect diffuse symptoms (mosaic and leaf-curl viruses, mites), and the faithfulness metrics in Table III depend on them. Evaluation against hand-drawn masks such as PlantSeg [20] is required for pixel-level claims.
2) Each configuration was trained once with one seed. A5 stopped early (epoch 10 of 18). Confidence intervals capture sampling variability of the test set, not training variability.
3) The CNN baselines were trained on a different split and may have seen test images. This biases their clean accuracy upwards; the like-for-like A0 baseline addresses it.
4) Generalisation to field images remains unsolved for all models (Table IV).

Future work. Fine-tuning on the PlantDoc training split with pseudo-masks generated by the same pipeline, so that adaptation speed can be compared across models; multi-leaf field segmentation; the remaining ablations with three seeds each; and evaluation on hand-annotated lesion masks.

VII. DEPLOYMENT IN AGRI-X

LesionNet is the default diagnostic model of AGRI-X, an open-source Streamlit application for Indian farmers (Fig. 6). One forward pass returns the class, the lesion map, the diseased share of the leaf and per-spot statistics. The severity grade and the disease drive treatment guidance, and a weather-based spread-risk score uses live forecasts. The application also provides location-aware crop recommendations, yield estimates and a farmer assistant, and a model-comparison page reproduces the robustness results of this paper. Because the displayed map is the classifier's actual evidence, a user who sees red on the table rather than on the leaf has a direct reason to distrust the result.

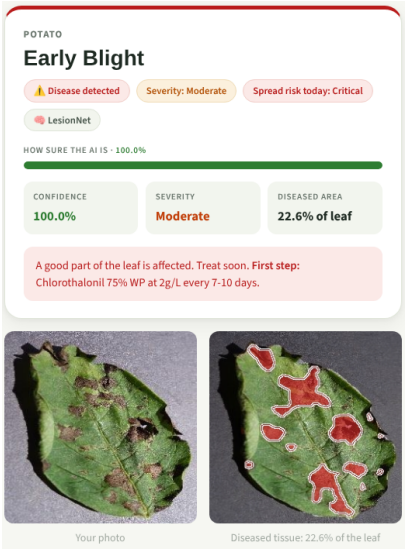


Fig. 6. LesionNet in the AGRI-X application: the diagnosis, the confidence, a severity grade from the predicted diseased area (mild $< 10\%$, moderate 10–25%, severe $> 25\%$) and the lesion map used for the decision.

VIII. Conclusion

High accuracy on PlantVillage does not imply that a model looks at the disease. Four standard CNNs place most of their evidence on healthy tissue and background, and the most accurate of them loses more than half its accuracy when the background is changed. LesionNet constrains the classifier to see only what its segmentation head outlines as leaf and lesion. Trained without manual annotation, it matches the best CNN on clean images, is nearly invariant to background replacement, yields a faithful lesion map by construction, and measures severity. Field images remain an open problem for every model studied, and we hope the released code, intervention protocol and negative results help the community measure progress on it honestly.

Data and Code Availability

Code, trained LesionNet weights, evaluation scripts and all result files are available at https://github.com/raj06872436/AI_Farming_Project (directory `research/lesion_net`). PlantVillage and PlantDoc are public datasets.

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